

Algebraic Manipulation

Worksheet 1 (Non-Calculator)

1. W14/qp11/q12

$$s = \frac{n}{2}(a + b)$$

- Evaluate s when $n = 200$, $a = 3.6$ and $b = 5.7$. [1]
- Rearrange the formula to make b the subject. [2]

2. S15/qp12/q15

$$c = \sqrt{8a - 3b}$$

- Find c when $a = 3$ and $b = -4$. [1]
- Rearrange the formula to make b the subject. [2]

3. W15/qp11/q15

$$4 = \sqrt{\frac{cx + 1}{dx - 1}}$$

Find x in terms of c and d . [3]

4. S16/qp12/q26(a)

Make p the subject of the formula $t = \frac{p+3}{p-4}$. [3]

5. S17/qp11/q16

- Given that $a = 3$ and $b = -7$, evaluate
 - $2a - b$ [1]
 - $a^2 + b^2$ [1]
- $A = 2r^2 + 5$. Rearrange the formula to make r the subject. [2]

6. S18/qp11/q22

$$s = \sqrt[3]{t + 4}$$

- Find s when $t = 121$. [1]
- Rearrange the formula to make t the subject. [2]

7. S19/qp11/q16

$$c = \frac{3b^2 - 2a}{5}$$

Rearrange the formula to make b the subject.

[3]

8. S19/qp12/q18

$$r = \frac{4p + 2}{3 - p}$$

a. Find r when $p = -2$

[1]

b. Rearrange the formula to make p the subject.

[3]

9. S20/qp12/q11

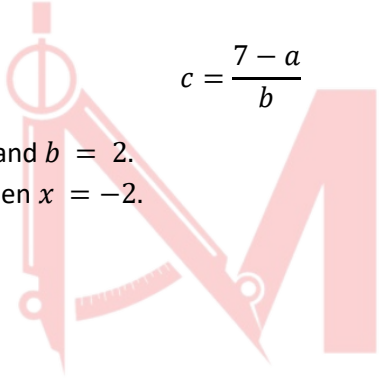
$$c = \frac{7 - a}{b}$$

a. Find c when $a = -4$ and $b = 2$.

[1]

b. $y = 5^x + 1$. Find y when $x = -2$.

[1]



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Answer Key:

1. (a) 930 (b) $\frac{2s-an}{n}$

2. (a) 6 (b) $\frac{8a-c^2}{3}$

3. $\frac{17}{16d-c}$

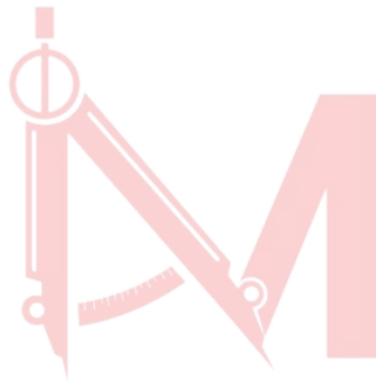
4. $\frac{3+4t}{t-1}$

5. (a)(i) 13 (ii) 58 (b) $r = \pm \sqrt{\frac{A-5}{2}}$

6. (a) 5 (b) $t = s^3 - 4$

7. $\sqrt{\frac{5c+2a}{3}}$

8. (a) -1.2 (b) $1\frac{1}{25}$



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